

IOT BASED SMART AGRICULTURE MONITORING SYSTEM

A mini project report submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD

In partial fulfillment of the requirement
for the award of the degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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SREE CHAITANYA INSTITUTE OF TECHNOLOGICAL SCIENCES
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CERTIFICATE

This is to certify that the mini project report entitled **“IOT BASED SMART AGRICULTURE MONITORING SYSTEM”** is being submitted by **BOGGULA NANDINI (23TR1A0407) MANDA ABHINAY (23TR1A0436) NAMPELLY KIRITI (23TR1A0442) PENCHALA SAIKUMAR (23TR1A0445)**, in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Technology in Electronics and Communication Engineering**, to the **Sree Chaitanya Institute Of Technological Sciences**, karimnagar, is a bonafide work carried out by them under my guidance and supervision

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ACKNOWLEDGEMENTS

The Satisfaction that accomplishes the successful completion of any task would be incomplete without the mention of the people who make it possible and whose constant guidance and encouragement crown all the efforts with success.

*It is our privilege and pleasure to express our profound sense of respect, gratitude and indebtedness to our guide **Mrs. M. SWAPNA**, Assistant professor, Department of ECE, SCITS, for her constant guidance, inspiration, and constant encouragement throughout this Mini project work,*

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*We sincerely extend our thanks to **Dr. A. PRASAD RAJU**, Principal, SCITS, Karimnagar, for providing all the facilities required for completion of this project.*

We would like to thank all the staff and all our friends for their good wishes, their helping hand and constructive criticism, which led the successful completion of this Mini project.

We are immensely indebted to our parents, brothers and sisters for their love and unshakable belief in us and the understanding and ever-decreasing grudges for not spending time more often. We will now, since the excuse is in the process of vanishing by being printed on these very pages.

Finally, we thank all those who directly and indirectly helped us in this regard. We apologize for not listing everyone here.

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DECLARATION

We here by declare that the work which is being presented in this dissertation entitled, “**IOT BASED SMART AGRICULTURE MONITORING SYSTEM**”, submitted towards the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering**, SCITS, Karimnagar, is an authentic record of our own work carried out under the guidance of **Mrs.M.SWAPNA**, Assistant professor, Department of ECE, SCITS, Karimnagar.

To the best of our knowledge and belief, this mini project bears no resemblance with any report submitted to SCITS or any other University for the award of any degree or diploma.

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ABSTRACT

In smart villages, access to sustainable energy services acts as a catalyst for development, enabling facility of internet connection for the new possibilities of increasing agricultural cultivation with proper information and guidance, access to clean water, sanitation and nutrition, the growth of productive enterprises to boost farmer's income. The development of a country depends on the village's development. Most of the agriculture productivity suffer greatly with unforeseen change in climate. Therefore, farmers need to get appropriate information's if any sudden climatic disruption occurs, it should notify on time to avoid any major damage in agricultural field.

As part of the smart village concept, an intelligent system is designed that may help a farmer to get basic facilities/infrastructure by agricultural development. Here an intelligent system is proposed on the fact of farmers getting all relevant details about the improvement in fertilization of soil and agriculture by delivering climate change information's through an IoT (Internet of Things) devices. This information's could be handled through website and mobile phones. To ease for farmer understandings all the facts and information related to soil moisture and climatic alerts are delivered as of their interest. This system may help its members to collaborate and take it to another level of requirement in improving their production capacity.

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CHAPTER 1

INTRODUCTION

1.1 PROBLEM DEFINITION

Agriculture is one of the most important sectors in India and plays a major role in the country's economy. A large portion of the population depends on agriculture for their livelihood. With the increasing population, the demand for food production is also rising rapidly. To meet this demand, it is necessary to adopt modern technologies that can improve agricultural productivity and efficiency.

In recent years, the integration of technology in agriculture has led to the concept of **Smart Agriculture**. Smart agriculture uses advanced technologies such as sensors, microcontrollers, and communication systems to monitor and manage farming activities efficiently. One of the most important technologies used in this field is the Internet of Things (IoT). IoT enables devices to communicate with each other and share data over the internet, making it possible to monitor agricultural fields remotely.

Traditional farming methods rely heavily on manual observation and control, which is time-consuming and inefficient. Farmers often face difficulties in maintaining proper soil moisture levels, monitoring temperature and humidity, and ensuring timely irrigation. These factors directly affect crop growth and yield. Lack of real-time information may lead to improper irrigation, resulting in either excess water usage or insufficient watering of crops.

The Smart Agriculture Monitoring System using IoT is designed to overcome these challenges. The system uses sensors such as soil moisture sensors and temperature-humidity sensors to continuously monitor environmental conditions. The collected data is processed using a microcontroller and used to control irrigation automatically. The system can also transmit data to the user through wireless communication, allowing remote monitoring and control.

1.2 OBJECTIVE OF PROJECT

The main objective of this project is to develop a Smart Agriculture Monitoring System using IoT that can improve farming efficiency and optimize resource usage. The system is designed to provide real-time monitoring and automatic control of irrigation based on environmental conditions.

specific objectives of the project are as follows:

- To measure soil moisture levels using a soil moisture sensor
- To monitor temperature and humidity using a DHT11 sensor
- To automate the irrigation system using a relay-controlled water pump
- To display real-time data on an LCD screen
- To transmit data to users through Wi-Fi using the ESP8266 module
- To reduce water wastage by providing controlled irrigation
- To minimize manual effort and improve agricultural productivity
- To develop a low-cost and efficient system suitable for farmers

1.3 ORGANIZATION OF THE DOCUMENTATION

This project report is organized into different chapters to provide a clear understanding of the system and its implementation. Each chapter focuses on a specific aspect of the project.

Chapter1: Introduction

This chapter provides an overview of the project, including the introduction, problem definition, objectives, and structure of the report.

Chapter2: Literature Survey

This chapter presents a review of existing systems and research work related to smart agriculture using IoT technology. It highlights the advantages and limitations of current solutions.

Chapter3: Design and Implementation

This chapter explains the overall system design, hardware and software requirements, block diagram, algorithm, and working of the proposed system.

Chapter4: Result Analysis

This chapter discusses the implementation results, system performance, output observations, and analysis of the developed system.

Chapter5: Conclusion

This chapter summarizes the project, discusses future enhancements, and lists possible applications of the system.

1.4 NEED FOR THE PROJECT

Agriculture today faces several challenges such as water scarcity, climate change, and increasing demand for food. Traditional farming methods are not sufficient to handle these challenges efficiently. There is a need for a smart system that can monitor environmental conditions accurately and provide proper irrigation at the right time. An automated system helps in reducing human effort and ensures better utilization of resources. By using IoT technology, farmers can monitor their fields remotely and make better decisions based on real-time data. This leads to improved crop growth and higher productivity.

1.5 SCOPE OF THE PROJECT

The Smart Agriculture Monitoring System is designed to provide efficient irrigation and monitoring for agricultural fields. The system focuses on small and medium-scale farming and helps farmers manage water usage effectively. The project can be extended by adding more sensors and advanced technologies. It can also be integrated with mobile applications and cloud platforms for better monitoring and control. The system is scalable and can be used in different agricultural environments.

1.6 LIMITATIONS

Although the system provides many advantages, it has some limitations:

- It depends on internet connectivity for remote monitoring
- Sensor readings may vary due to environmental conditions
- Initial setup cost may be required
- Limited to small-scale implementation without expansion

1.7 APPLICATIONS

The proposed system can be used in various areas such as:

- Agricultural fields for automatic irrigation
- Greenhouses for controlled environment monitoring
- Gardens and nurseries
- Smart farming systems
- Water resource management

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Agriculture is one of the most important sectors contributing to the economy and food security of a country. However, traditional agricultural practices are no longer sufficient to meet the increasing demand for food production. Farmers face multiple challenges such as unpredictable climate conditions, water scarcity, and lack of proper monitoring systems. These challenges directly affect crop yield and productivity. With the advancement of technology, agriculture is gradually shifting towards automation and smart systems. One of the key technologies enabling this transformation is the Internet of Things (IoT). IoT allows devices such as sensors and controllers to communicate with each other and exchange data through the internet. In agriculture, IoT is used to monitor soil moisture, temperature, humidity, and other environmental parameters in real-time.

Many researchers have proposed IoT-based smart agriculture systems to improve farming efficiency. These systems use sensors to collect data and microcontrollers to process the data and take appropriate actions such as controlling irrigation. The collected data can also be transmitted to farmers through mobile applications or web platforms, enabling remote monitoring.

Despite the availability of various systems, there are still limitations in terms of cost, complexity, and efficiency. Therefore, it is necessary to study existing systems and identify their drawbacks in order to design an improved solution. This chapter presents an overview of existing systems, their disadvantages, and the advantages of the proposed system.

2.2 EXISTING SYSTEM

Traditional agriculture systems mainly depend on manual methods for irrigation and monitoring. Farmers usually make decisions based on experience rather than actual data. Irrigation is often performed at fixed time intervals without considering the actual moisture level of the soil. This leads to inefficient use of water and may negatively impact crop growth.

Some modern systems have introduced basic automation techniques such as timer-based irrigation systems. These systems supply water at predefined intervals, reducing manual effort to some extent. However, they do not consider real-time environmental conditions, making them unreliable in varying climates.

Sensor-based systems have also been developed to monitor parameters such as soil moisture, temperature, and humidity. These systems provide more accurate data compared to traditional methods. However, many of these systems lack proper integration with communication technologies, which limits remote access and monitoring.

In addition, some IoT-based systems have been proposed that use wireless sensor networks for data collection. Although these systems improve monitoring capabilities, they often require complex setup, continuous internet connectivity, and higher costs. As a result, they are not easily accessible to all farmers.

Thus, existing systems provide partial solutions but fail to deliver a complete, efficient, and user-friendly system for smart agriculture.

2.3 DISADVANTAGES OF EXISTING SYSTEM

The existing agriculture systems suffer from several limitations that reduce their effectiveness and reliability. One of the major drawbacks is the lack of real-time monitoring. Traditional and timer-based systems do not provide accurate information about soil moisture and environmental conditions, leading to improper irrigation practices.

Another significant disadvantage is inefficient water usage. Since irrigation is not based on actual soil conditions, water is often wasted or insufficiently supplied. This not only affects crop growth but also contributes to resource depletion. Most existing systems require manual intervention, which increases labor effort and time consumption. Farmers need to regularly visit their fields to monitor conditions and operate irrigation systems. This is not practical, especially for large agricultural lands. In many sensor-based systems, there is no provision for remote monitoring or control. Farmers cannot access real-time data from distant locations, which limits the usefulness of these systems.

Additionally, some IoT-based systems depend heavily on continuous internet connectivity, making them unreliable in rural areas with poor network coverage. High installation cost and system complexity are also major concerns. Advanced systems require technical knowledge for setup and maintenance, making them less suitable for small-scale farmers. These disadvantages highlight the need for a more efficient, cost-effective, and easy-to-use system.

2.4 PROPOSED SYSTEM

To overcome the limitations of existing systems, a Smart Agriculture Monitoring System using IoT is proposed. This system is designed to provide real-time monitoring of environmental parameters and automate irrigation based on actual field conditions. The system uses a soil moisture sensor to measure the water content in the soil. Based on the sensor readings, the system determines whether irrigation is required. If the soil is dry, the water pump is automatically turned ON using a relay. If sufficient moisture is detected, the pump is turned OFF. This ensures efficient water usage and prevents wastage.

A temperature and humidity sensor (DHT11) is used to monitor environmental conditions. The collected data is processed using a microcontroller (Arduino UNO), which acts as the central control unit of the system. The system also includes a Wi-Fi module (ESP8266) to transmit data to users, enabling remote monitoring.

Additionally, an LCD display is used to show real-time sensor readings locally. The system is designed to be simple, cost-effective, and easy to use, making it suitable for farmers. By integrating sensing, processing, and communication technologies, the proposed system provides a complete solution for smart agriculture.

2.5 ADVANTAGES OF PROPOSED SYSTEM

The proposed system offers several advantages over traditional and existing agriculture systems. One of the main advantages is real-time monitoring of soil and environmental conditions. This allows farmers to make informed decisions and take timely actions. Another important advantage is automatic irrigation control. The system operates based on actual soil moisture levels, ensuring efficient water usage. This reduces water wastage and improves crop growth.

The system also provides remote monitoring capabilities through IoT technology. Farmers can access data from anywhere using mobile devices or computers. This reduces the need for constant physical presence in the field.

In addition, the system reduces manual labor and saves time. Automation simplifies agricultural operations and increases overall efficiency. The system is also cost-effective and easy to implement, making it suitable for small and medium-scale farmers. Overall, the proposed system improves productivity, conserves resources, and enhances the efficiency of agricultural practice

2.6 SMART AGRICULTURE SYSTEMS

Smart agriculture systems represent a major advancement in the field of modern farming by integrating technology into traditional agricultural practices. These systems are designed to monitor, analyze, and control various environmental parameters in real time, allowing farmers to make informed decisions. In the proposed system, sensors such as soil moisture sensors and temperature-humidity sensors are used to collect data continuously from the field. This data is then processed by a microcontroller, which acts as the brain of the system and controls the entire operation.

One of the key features of smart agriculture systems is their ability to provide continuous monitoring without the need for human intervention. Unlike traditional farming methods, where farmers must manually check field conditions, smart systems operate automatically and provide accurate information at all times. This reduces dependency on human effort and increases the reliability of agricultural operations.

Another important advantage is efficient resource management. Water, which is one of the most critical resources in agriculture, is used more effectively in smart systems. Instead of supplying water at fixed intervals, the system provides water only when required based on real-time soil moisture data. This not only reduces wastage but also improves crop health. In addition, smart agriculture systems are flexible and scalable. They can be expanded by adding more sensors or integrating additional technologies such as cloud computing and data analytics. This makes them suitable for both small-scale and large-scale farming. The ability to adapt to different agricultural conditions makes these systems highly practical.

Furthermore, smart agriculture contributes to sustainable farming practices. By optimizing resource usage and reducing manual errors, these systems help in maintaining soil health and improving productivity. As technology continues to evolve, smart agriculture systems are expected to become more advanced and widely adopted across the world.

2.7 AUTOMATION IN IRRIGATION SYSTEMS

Automation plays a crucial role in improving the efficiency and accuracy of irrigation systems. Traditional irrigation methods rely heavily on manual operation, where farmers decide when and how much water to supply. This approach often leads to improper irrigation, either supplying excess water or insufficient water, both of which can negatively affect crop growth.

Automated irrigation systems overcome these limitations by using sensors and control mechanisms. In the proposed system, the soil moisture sensor continuously monitors the moisture level in the soil. When the moisture level drops below a predefined threshold, the microcontroller sends a signal to activate the water pump through a relay module. This ensures that irrigation is performed only when necessary.

As the soil absorbs water and the moisture level increases, the system keeps monitoring the changes. Once the required moisture level is reached, the system automatically turns OFF the pump. This prevents over-irrigation and ensures optimal water usage. The entire process is carried out automatically without any human intervention.

Another important benefit of automation is consistency. Unlike manual methods, where decisions may vary based on human judgment, automated systems follow a fixed logic, ensuring uniform operation. This improves the overall efficiency of irrigation and supports better crop management.

Automation also helps in saving time and labor. Farmers do not need to constantly monitor field conditions or operate irrigation systems manually. This allows them to focus on other important agricultural activities. In addition, automated systems reduce energy consumption by operating the pump only when required.

2.8 ROLE OF IOT IN MODERN AGRICULTURE

The Internet of Things (IoT) has revolutionized the field of agriculture by enabling real-time communication between devices. IoT allows sensors, microcontrollers, and communication modules to work together as a connected system. In the proposed system, IoT is used to transmit sensor data to users, enabling remote monitoring.

The Wi-Fi module plays a key role in data transmission. It sends the collected sensor data to an online platform, where it can be accessed by farmers using mobile devices or computers. This allows users to monitor field conditions from anywhere, reducing the need for physical presence.

One of the major advantages of IoT is real-time monitoring. Farmers can observe changes in soil moisture, temperature, and humidity instantly and take appropriate actions. This improves decision-making and ensures timely response to environmental changes.

IoT also supports data storage and analysis. The data collected over time can be stored and used to identify patterns and trends. This helps in predicting irrigation requirements and improving farming strategies. With the help of data analysis, farmers can optimize resource usage and increase productivity.

Another important feature of IoT is its ability to integrate multiple systems into a single platform. This creates a smart and connected environment where all components work together efficiently. IoT-based systems are flexible and can be upgraded with additional features such as mobile alerts and cloud storage.

Thus, IoT plays a vital role in transforming traditional agriculture into a modern, efficient, and intelligent system.

2.9 CHALLENGES IN EXISTING AGRICULTURE SYSTEMS

Despite technological advancements, many existing agriculture systems still face several challenges. One of the major issues is the lack of real-time monitoring in traditional farming methods. Farmers often rely on experience and manual observation, which may not always be accurate. This can lead to improper irrigation and reduced crop yield.

Water management is another significant challenge. Many systems do not use water efficiently, resulting in wastage or insufficient supply. Over-irrigation can damage crops and affect soil quality, while under-irrigation can hinder plant growth. In addition, the cost of advanced agricultural systems can be high, making them inaccessible to small-scale farmers. Complex systems require technical knowledge for installation and maintenance, which further limits their adoption.

Connectivity issues also affect the performance of IoT-based systems, especially in rural areas where internet access may be limited. Without stable connectivity, remote monitoring becomes difficult. Power supply is another concern, as continuous operation of sensors and communication modules requires reliable energy sources. In areas with frequent power cuts, system performance may be affected. These challenges highlight the need for simple, cost-effective, and efficient solutions that can be easily implemented in real-world agricultural environments.

2.10 NEED FOR SMART AGRICULTURE SOLUTIONS

The increasing demand for food production, along with limited natural resources, has created a strong need for smart agriculture solutions. Traditional farming methods are not capable of meeting these demands efficiently. Smart systems provide a modern approach by using technology to improve productivity and resource management. The proposed system addresses these challenges by providing real-time monitoring and automatic irrigation. It ensures that crops receive the right amount of water based on actual soil conditions. This improves crop health and increases yield.

Smart agriculture solutions also promote sustainable farming practices. By reducing water wastage and optimizing resource usage, these systems help in conserving natural resources. This is essential for long-term agricultural development. Another important benefit is improved decision-making. Farmers can rely on accurate data instead of assumptions, leading to better results. The use of technology also reduces manual effort and increases efficiency. In addition, smart systems can be expanded and upgraded with advanced technologies such as Artificial Intelligence and data analytics. This allows further improvement in agricultural practices. Therefore, the adoption of smart agriculture solutions is essential for improving productivity, ensuring sustainability, and meeting future agricultural demands.

CHAPTER 3

DESIGN AND IMPLEMENTATION

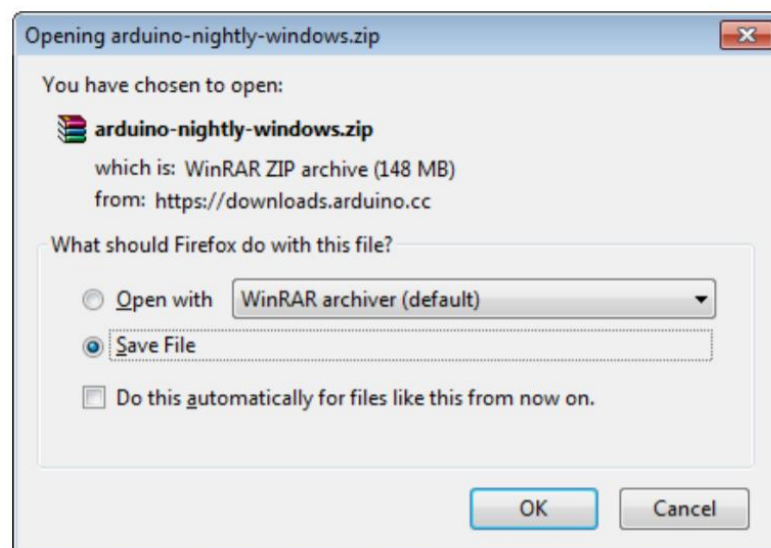
3.1 INTRODUCTION

This chapter presents the design and implementation of the Smart Agriculture Monitoring System using IoT. The system is developed to monitor environmental parameters such as soil moisture, temperature, and humidity, and to automate irrigation based on real-time conditions. The main objective of the system is to improve agricultural efficiency by reducing manual effort and ensuring proper water management. The system uses sensors to collect data from the field and a microcontroller to process the data. Based on the processed data, the system automatically controls the water pump using a relay module. The integration of IoT technology enables remote monitoring of the system, allowing farmers to access real-time information from anywhere. The design of the system focuses on simplicity, low cost, and reliability so that it can be easily used in agricultural applications.

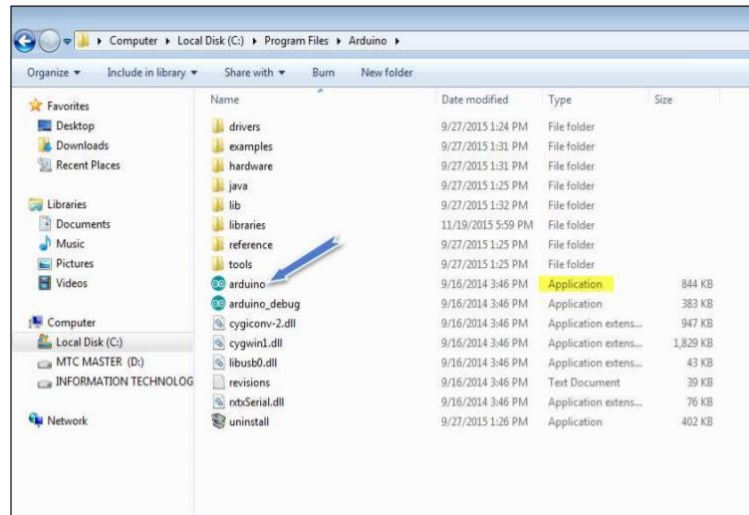
3.2 SOFTWARE REQUIREMENT SPECIFICATION

1) ARDUINO COMPILER (IDE) :

Arduino IDE Software. You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.



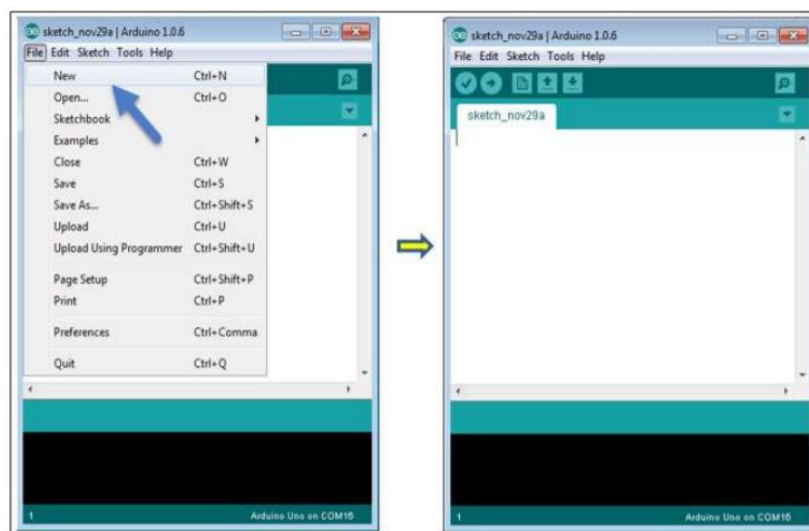
Launch Arduino IDE. After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Doubleclick the icon to start the IDE.

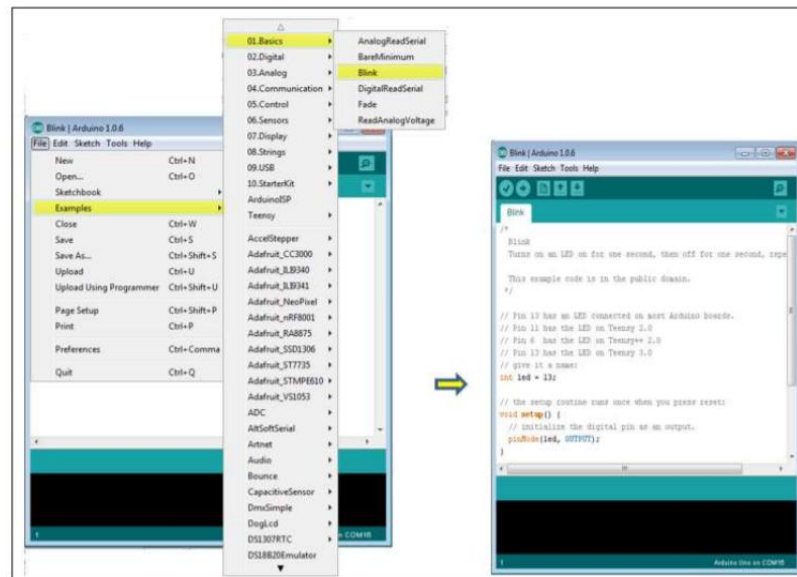


Open your first project. Once the software starts, you have two options:

- Create a new project.
- Open an existing project example.

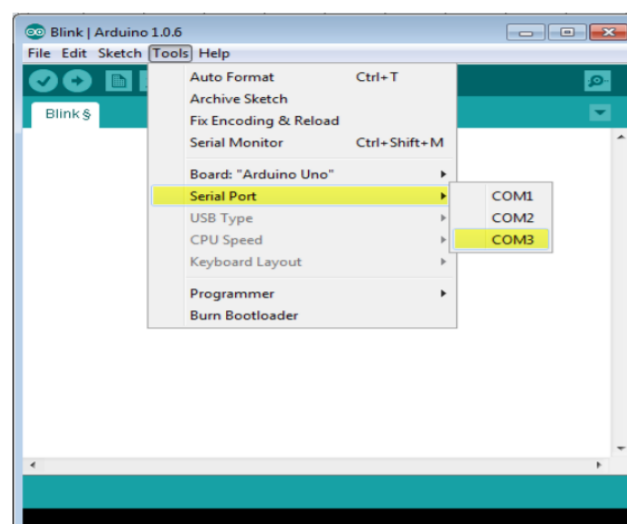
To create a new project, select File --> New





Here, we are selecting just one of the examples with the name Blink. It turns the LED on and off with some time delay. You can select any other example from the list

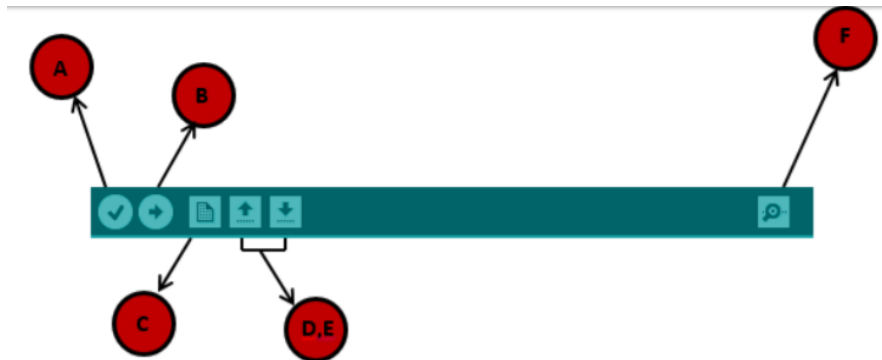
Select your serial port. Select the serial device of the Arduino board. Go to Tools -> Serial Port menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.



Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar.

A- Used to check if there is any compilation error.

- B- Used to upload a program to the Arduino board.
- C- Shortcut used to create a new sketch.
- D- Used to directly open one of the example sketch.
- E- Used to save your sketch.

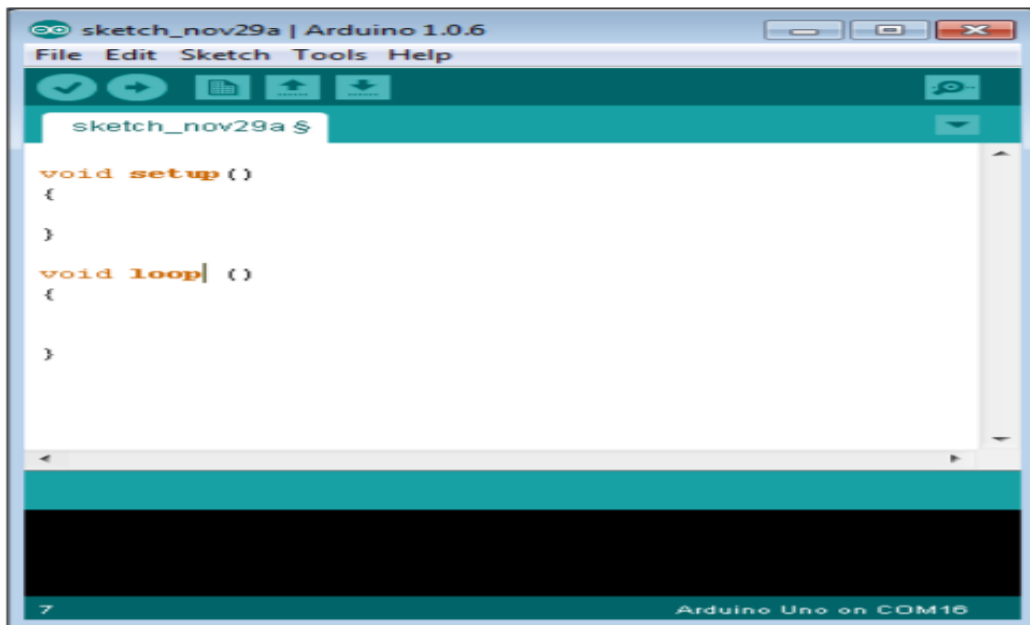


In this chapter, we will study in depth, the Arduino program structure and we will learn more new terminologies used in the Arduino world. The Arduino software is open-source. The source code for the Java environment is released under the GPL and the C/C++ microcontroller libraries are under the LGPL. Sketch: The first new terminology is the Arduino program called “sketch”. Structure Arduino programs can be divided in three main parts: Structure, Values, and Functions.

In this tutorial, we will learn about the Arduino software program, step by step, and how we can write the program without any syntax or compilation error. Let us start with the Structure.

Software structure consist of two main functions:

- Setup() function
- Loop() function



Data types in C refers to an extensive system used for declaring variables or functions of different types. The type of a variable determines how much space it occupies in the storage and how the bit pattern stored is interpreted. The following table provides all the data types that you will use during Arduino programming.

2) THINGSPEAK

ThingSpeak is an open-source IoT (Internet of Things) analytics platform that enables users to collect, analyze, and visualize sensor data in real time. It serves as a bridge between IoT devices and the cloud, allowing seamless data communication over the internet. Developed by MathWorks, ThingSpeak is widely used by students, researchers, and developers for building smart applications, especially in fields like home automation, agriculture, weather monitoring, and industrial automation.

How ThingSpeak Works

Data Collection: Sensors or devices like ESP8266/NodeMCU collect environmental or physical data (e.g., temperature, humidity).

Data Transmission: The collected data is sent to ThingSpeak via Wi-Fi using HTTP POST requests to the channel's API key.

Data Storage: The data is securely stored in fields within a ThingSpeak channel. Each update is timestamped.

Data Visualization: The stored data is displayed in real-time graphs or charts on the ThingSpeak dashboard.

Data Analysis: MATLAB scripts can be written to analyze data patterns, detect anomalies, or perform calculations.

Trigger Actions: Based on the analyzed results, ThingSpeak can send alerts or activate other devices/services using IFTTT or webhooks.

ThingSpeak as a Cloud Interface

ThingSpeak represents itself as a reliable and flexible **cloud interface** that bridges the gap between IoT devices and online data services. Acting as a real-time cloud platform, ThingSpeak enables devices to send data over the internet and store it securely in the cloud, where it can be accessed, monitored, and analyzed from anywhere in the world.

As a cloud interface, ThingSpeak provides several key functionalities:

Remote Data Logging: Sensor data from microcontrollers is sent to ThingSpeak, where it is time-stamped and stored in the cloud, eliminating the need for local storage.

Live Data Visualization: Users can instantly visualize incoming data through dynamic graphs, gauges, and dashboards, making it easier to monitor trends and system behaviour.

Data Sharing and Collaboration: ThingSpeak supports both public and private channels, enabling users to share real-time sensor data with others securely or openly.

Cloud-Based Analytics: Through built-in MATLAB integration, users can process and analyze data directly on the cloud without the need for external software.

IoT Interfacing: ThingSpeak integrates well with services like IFTTT and webhooks, allowing cloud-based automation and interactions with other online platforms or devices.

In essence, ThingSpeak provides a robust and user-friendly cloud interface that supports the complete data cycle—from acquisition and storage to visualization and intelligent decision-making. It exemplifies the core principles of IoT by making devices "smart" through seamless internet connectivity and cloud-based intelligence

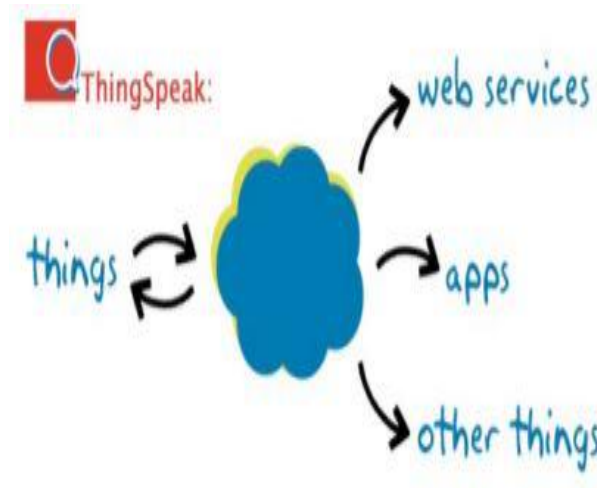


Fig 3.1: Thingspeak

3.3 User Requirements

The system is designed based on the needs of farmers to make irrigation efficient and easy.

Functional Requirements:

- Monitor soil moisture level
- Measure temperature and humidity
- Automatically control water pump
- Display data on LCD
- Send data through Wi-Fi

Non-Functional Requirements:

- Low cost and easy to use
- Low power consumption
- Minimal maintenance

3.4 ALGORITHM

The working of the system is explained using the following algorithm:

1. Start the system
2. Initialize all components
3. Read soil moisture value
4. Read temperature and humidity values
5. Compare soil moisture with threshold value
6. If soil is dry → Turn ON motor
7. Else → Turn OFF motor
8. Display values on LCD
9. Send data via Wi-Fi module
10. Repeat continuously

3.5 WORKING PRINCIPLE

The Smart Agriculture Monitoring System operates based on real-time sensor data and automatic control. The soil moisture sensor continuously measures the moisture level in the soil and sends the data to the Arduino controller.

The Arduino compares this value with a predefined threshold. If the soil moisture is below the threshold, the system activates the relay module, which turns ON the water pump. When the soil reaches the required moisture level, the pump is turned OFF automatically.

The DHT11 sensor measures temperature and humidity, providing additional environmental data. The LCD displays real-time values, and the Wi-Fi module transmits the data to the user.

3.6 HARDWARE REQUIREMENTS

1. ARDUINO UNO
2. DHT11
3. RELAY
4. SOIL MOSITURE
5. LCD
6. PUMP
7. WI-FI

3.7 BLOCK DIAGRAM

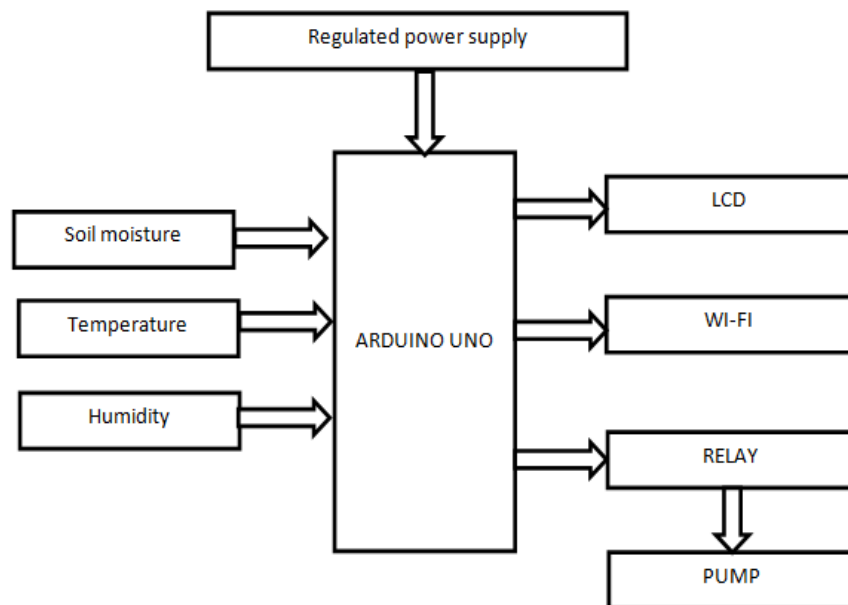


Fig. 3.2 Block Diagram

1. ARDUINO UNO

The most common version of Arduino is the Arduino Uno. This board is what most people are talking about when they refer to an Arduino. The Uno is one of the more popular boards in the Arduino family and a great choice for beginners. There are different revisions of Arduino Uno, below detail is the most recent revision (Rev3 or R3). The Arduino Uno is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button.

It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started.

Features

Microcontroller	:	ATmega328
Operating Voltage	:	5V
Input Voltage (recommended)	:	7-12V
Input Voltage (limits)	:	6-20V
Digital I/O Pins	:	14 (of which 6 provide PWM output)
Analog Input Pins	:	6
DC Current per I/O Pin	:	40 mA
DC Current for 3.3V Pin	:	50 mA
Flash Memory	:	32 KB (ATmega328) of which 0.5 KB used by boot loader
SRAM	:	2 KB (ATmega328)
EEPROM	:	1 KB (ATmega328)
Clock Speed	:	16 MHz
Length	:	68.6 mm
Width	:	53.4 mm

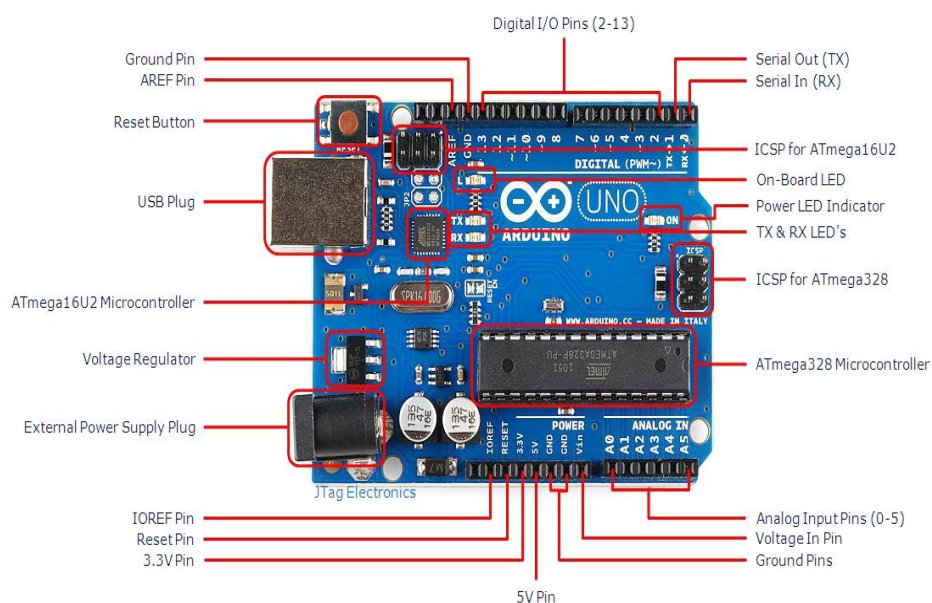


Fig. 3.3 Arduino Board

Arduino Uno Board

➤ USB Plug & External Power Supply

The Arduino Uno can be powered either through a USB cable connected to a computer or using an external power supply via a barrel jack. The board automatically selects the power source. The USB connection is also used to upload programs. The recommended external voltage range is 7V to 12V. Supplying less than 7V may cause unstable operation, while more than 12V can overheat and damage the board.

➤ Voltage Regulator

The voltage regulator controls the voltage supplied to the Arduino board and protects it from excessive voltage. It ensures a stable 5V output for proper functioning. However, it has limits, so the input voltage should not exceed safe levels.

➤ Power Pins

- VIN: Used to supply external voltage (7–12V)
- 5V: Provides regulated 5V output to components
- 3.3V: Provides 3.3V output (max 50mA)

➤ Ground (GND) Pins

Multiple ground pins are available and are used to complete the circuit by providing a common ground connection.

➤ IOREF Pin

This pin provides the reference voltage at which the microcontroller operates. It helps connected devices (shields) adapt to the correct voltage level.

➤ Digital Input/Output Pins (D0–D13)

The Arduino Uno has 14 digital pins, which can be used as input or output. Each pin operates at 5V and can handle up to 40mA. Some pins (3, 5, 6, 9, 10, 11) support PWM, allowing analog-like output.

➤ Serial Communication (TX & RX)

- TX (D1): Transmits data
- RX(D0):Receivesdata.These pins are used for communication with devices like Wi-Fi or GSM modules.

➤ **External Interrupt Pins**

Pins 2 and 3 support interrupts, allowing the Arduino to respond immediately to external signals like vibration detection.

➤ **SPI Communication**

Pins 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK) are used for SPI communication with other devices.

➤ **Analog Input Pins (A0–A5)**

These pins read analog signals (0–5V) and convert them into digital values (0–1023). They are used for sensors like temperature, gas, and water level sensors. Pins A4 (SDA) and A5 (SCL) also support I2C communication.

➤ **Reset Pin**

The reset pin restarts the microcontroller and begins program execution from the start.

LED Indicators

- Power LED: Indicates that the board is powered
- Built-in LED (Pin 13): Used for testing programs
- TX/RX LEDs: Indicate data transmission and reception

➤ **Reset Button**

Pressing the reset button restarts the Arduino program, which is useful for testing and debugging.

2. DHT11

The DHT11 sensor is an ultralow-cost temperature and humidity sensor. It uses a capacitance dampness sensor and a thermostat to calculate nearby surrounding, and throws out an output signal on the data pin. It can be used very easily, but needs correct timing to achieve data. Further, the sensor operates with minimal delay in updating values, which helps the system maintain accurate environmental monitoring. The continuous flow of temperature and humidity data supports better observation of field conditions over time. This information can be useful in identifying patterns such as increased temperature leading to faster soil drying. The sensor works reliably within its operating range and contributes to the overall stability of the system. Its simple interfacing and consistent output make it a suitable choice for this project.

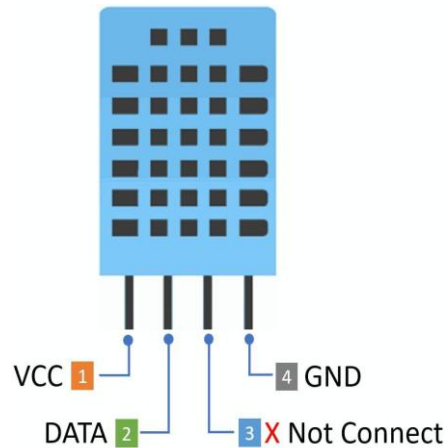


Fig. 3.4 DHT11

3. RELAY

A relay is a switch. This is an electromagnet switch in which it is operate as a mechanically and also other operating principles are also used, such as Solid-state relays. It can be used to control a circuit by a separate low-power signal, or must be controlled by one signal. The relays wereused in many fields. One of the fields used as an amplifier: they repeated the signal coming in from one circuit and retransmitted it on another circuit. It can be used extensively in telephone exchanges and early computers to perform logical operations. When an electric current is passed through the coil in which the coil is placed inside the relay and it generates a magnetic field. The relays are manufactured to operate quickly. It reduces the noise in low voltage application and it reduces arcing in high voltage application.

In addition to its switching function, the relay module maintains stable operation even during repeated ON and OFF cycles. This ensures that the irrigation system can function continuously without interruption. The relay also helps in maintaining proper electrical separation between control and power circuits, which improves safety. Its quick response time ensures that the pump reacts immediately to changes in soil conditions. This makes the overall system more responsive and dependable.



Fig. 3.5 Relay

4. SOIL MOSITURE:

Soil moisture is the water content of the soil. It can be expressed in terms of volume or weight. Soil moisture measurement can be based on *in situ* probes (e.g., capacitance probes, neutron probes) or remote sensing methods. Water that enters a field is removed from a field by runoff, drainage, evaporation or transpiration. Runoff is the water that flows on the surface to the edge of the field; drainage is the water that flows through the soil downward or toward the edge of the field underground; evaporative water loss from a field is that part of the water that evaporates into the atmosphere directly from the field's surface; transpiration is the loss of water from the field by its evaporation from the plant itself.

Further, the sensor provides continuous feedback to the system, allowing it to monitor even small changes in soil moisture levels. This helps in maintaining precise control over irrigation. The sensor ensures that water is supplied only when necessary, which improves efficiency. Its consistent readings support better system performance and reduce the chances of incorrect irrigation decisions. This makes it an essential component for achieving accurate automation.

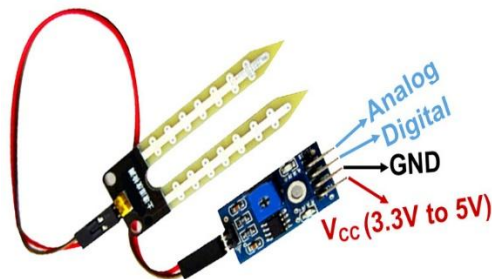


Fig. 3.6 Soil Moisture Sensor

5. LCD:

LCD (Liquid Crystal Display) Shown in fig. is an electronic display module. A 16x2 LCD display is very basic module and is very commonly used in different types of devices and circuits. These modules are preferred over seven segment and other multi segment LEDs.



Fig. 3.7 LCD Display

16x2 LCD

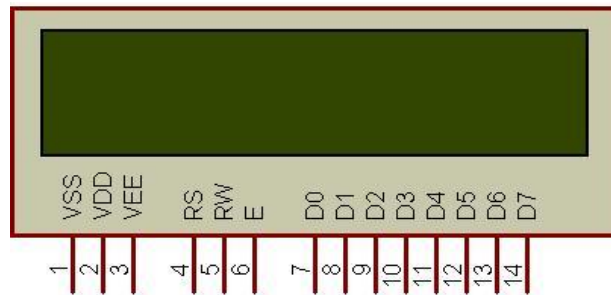
A 16x2 LCD means it display 16 characters per line and there are 2 lines. By this LCD each character is displayed in 5x7 pixel matrix. This LCD contain two registers, thats are Command and Data. The command register stores the command instructions which is given to the LCD. A command is an instruction which is given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD.

A **16x2 LCD** means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data.

The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD.



Fig 3.8 2x16 LCD Display



The 16x2 LCD display consists of 8 data pins (D0–D7) and control pins such as RS, R/W, and Enable. The data pins are used to send alphanumeric characters in ASCII format from the Arduino to the display. The RS (Register Select) pin is used to choose between command mode (RS = 0) and data mode (RS = 1). The R/W pin determines whether data is written to or read from the LCD, but it is usually set to write mode (0). The Enable (E) pin is used to latch the data, where a high-to-low signal is required to process the input. These pins work together to display information on the LCD screen.

6. PUMP:

A pump is a device that moves fluids (liquids or gases), or sometimes slurries, by mechanical action, typically converted from electrical energy into hydraulic energy. Mechanical pumps serve in a wide range of applications such as pumping water from wells, aquarium filtering, pond filtering and aeration, in the car industry for water-cooling and fuel injection, in the energy industry for pumping oil and natural gas or for operating cooling towers and other components of heating, ventilation and air conditioning systems. In the medical industry, pumps are used for biochemical processes in developing and manufacturing medicine, and as artificial replacements for body parts, in particular the artificial heart and penile prosthesis.

When a pump contains two or more pump mechanisms with fluid being directed to flow through them in series, it is called a multi-stage pump. Terms such as two-stage or double-stage may be used to specifically describe the number of stages. A pump that does not fit this description is simply a single-stage pump in contrast. The water pump functions as the main output device of the system, responsible for delivering water to the soil. It operates automatically based on the signals received from the relay module. When the soil moisture level falls below the required value, the pump is activated to supply water.

Once the desired moisture level is achieved, the pump stops operation. This controlled functioning ensures efficient use of water and reduces wastage. The pump works in coordination with other components to maintain proper irrigation conditions. Its automatic operation simplifies the irrigation process and improves overall system efficiency.



Fig. 3.9 Pump

7. WI-FI

Wi-Fi is a family of wireless network protocols, based on the IEEE 802.11 family of standards, which are commonly used for local area networking of devices and Internet access, allowing nearby digital devices to exchange data by radio waves. These are the most widely used computer networks in the world, used globally in home and small office networks to link desktop and laptop computers, tablet computers, smartphones, smart TVs, printers, and smart speakers together and to a wireless router to connect them to the Internet, and in wireless access points in public places like coffee shops, hotels, libraries and airports to provide visitors with Internet access for their mobile devices. Wi-Fi is a trademark of the non-profit Wi-Fi Alliance, which restricts the use of the term Wi-Fi Certified to products that successfully complete interoperability certification testing.

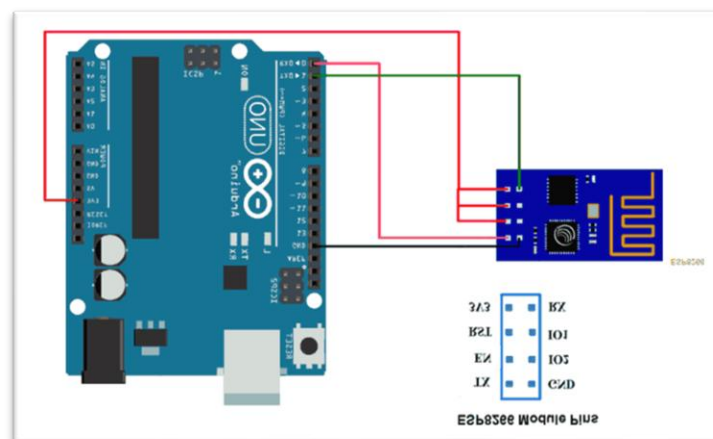
As of 2017, the Wi-Fi Alliance consisted of more than 800 companies from around the world.^[6] As of 2019, over 3.05 billion Wi-Fi enabled devices are shipped globally each year.

Wi-Fi uses multiple parts of the IEEE 802 protocol family and is designed to interwork seamlessly with its wired sibling, Ethernet. Compatible devices can network through wireless access points to each other as well as to wired devices and the Internet.

In addition to its basic function, the Wi-Fi module enables continuous data transmission between the system and the user. It ensures that the sensor data is updated regularly without interruption. This allows users to monitor field conditions in real time from remote locations. The module works efficiently when a stable internet connection is available, providing reliable communication between devices.

Furthermore, the Wi-Fi module supports easy integration with IoT platforms, making the system more flexible and advanced. It helps in reducing the need for physical monitoring by allowing users to access data through mobile devices or computers. The module also contributes to faster data transfer, ensuring that updates are received without delay.

The use of the Wi-Fi module enhances the overall system by adding remote monitoring capability. It improves convenience and makes the system more suitable for modern smart agriculture application



CHAPTER 4

RESULT ANALYSIS

4.1 INTRODUCTION

This chapter presents the results obtained from the implementation of the Smart Agriculture Monitoring System using IoT. The system was designed to monitor soil moisture, temperature, and humidity, and to control irrigation automatically based on real-time environmental conditions.

The purpose of this chapter is to evaluate the performance of the system under different conditions and to verify whether it meets the objectives defined in earlier chapters. The system was tested by placing sensors in soil under varying moisture levels and observing the response of the system.

The results obtained from the system include sensor readings, motor operation status, and data transmission through the Wi-Fi module. These results are analyzed to determine the efficiency, accuracy, and reliability of the system. The performance is evaluated based on how quickly and accurately the system responds to changes in soil moisture and environmental

4.2 EXPLANATION OF KEY FUNCTIONS

Soil Moisture Monitoring

The soil moisture sensor is used to measure the water content present in the soil. It continuously monitors the moisture level and sends the data to the Arduino controller. When the soil becomes dry, the sensor value decreases, indicating the need for irrigation.

Automatic Irrigation Control

The system automatically controls the water pump using a relay module. When the soil moisture level falls below a predefined threshold, the Arduino activates the relay, which turns ON the motor. Once the soil reaches the required moisture level, the motor is turned OFF automatically. This ensures efficient water usage and prevents over-irrigation.

Temperature and Humidity Monitoring

The DHT11 sensor is used to measure temperature and humidity of the surrounding environment. These parameters are important for understanding climatic conditions and their effect on crop growth. The data collected from the sensor helps in better monitoring of environmental conditions.

Display System

The LCD display shows real-time values of soil moisture, temperature, and humidity. This allows farmers to monitor the system locally without the need for additional devices. The display provides clear and continuous updates.

Wi-Fi Communication

The ESP8266 Wi-Fi module is used to transmit sensor data to the user. This enables remote monitoring of the agricultural field. Farmers can access the data using mobile devices or computers, reducing the need for physical presence in the field.

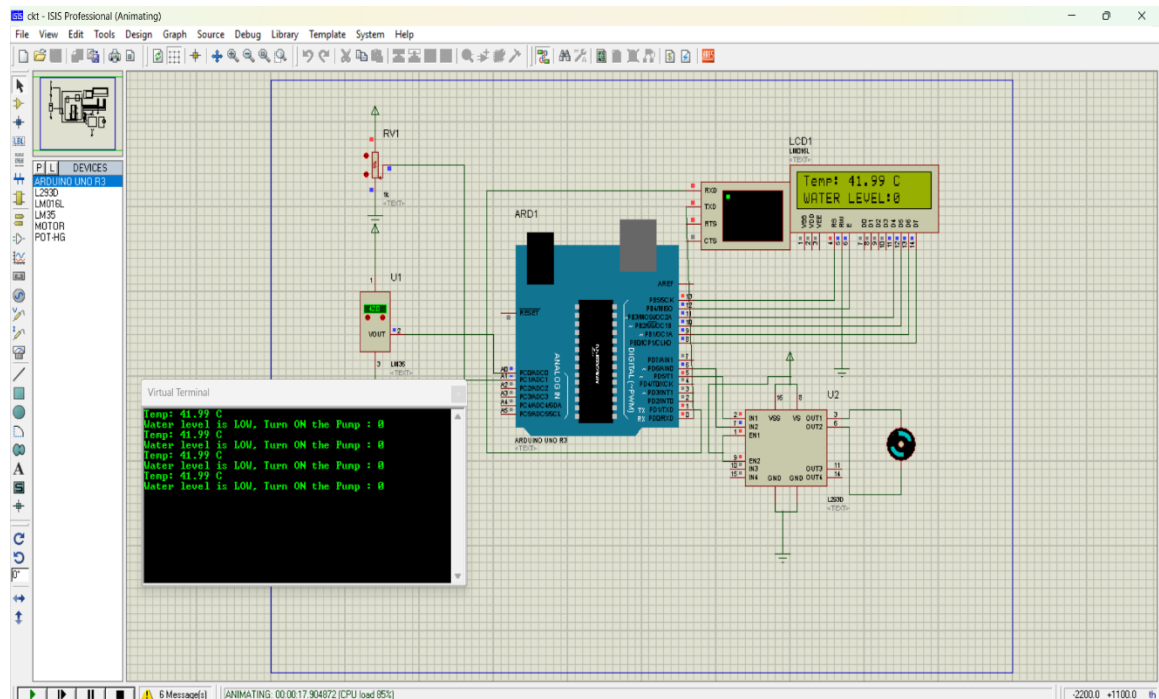
4.3 METHOD OF IMPLEMENTATION

The system is implemented using both hardware and software components. The soil moisture sensor and DHT11 sensor are connected to the input pins of the Arduino UNO. The relay module is connected to a digital output pin to control the water pump.

The Arduino is programmed using Embedded C in the Arduino IDE. The program continuously reads sensor values and compares them with predefined threshold values. Based on this comparison, the Arduino sends signals to the relay module to turn the motor ON or OFF.

The LCD display is interfaced with the Arduino to show real-time values of all parameters. The ESP8266 Wi-Fi module is connected to enable wireless communication. The data is transmitted to the user, allowing remote monitoring.

4.4 OUTPUT SCREEN



4.5 RESULT ANALYSIS

The system was tested under different soil moisture conditions to analyze its performance. When the soil moisture level was low, the system successfully detected the dry condition and activated the water pump. This ensured that water was supplied to the crops at the right time.

When the soil reached the required moisture level, the system automatically turned OFF the motor. This prevented over-irrigation and saved water. The transition between ON and OFF states of the motor was smooth and responsive.

The sensor readings were found to be accurate and consistent during testing. The system responded quickly to changes in environmental conditions, which indicates good performance. The LCD display provided clear and continuous updates, and the Wi-Fi module successfully transmitted data for remote monitoring.

The performance of the system was tested under different soil moisture conditions to verify its efficiency. When the soil was completely dry, the sensor detected low moisture levels and immediately activated the water pump. This ensured that water was supplied to the crops without delay.

When the soil reached a moderate moisture level, the system continued monitoring without activating the pump unnecessarily. Once the soil became sufficiently wet, the system automatically turned OFF the pump.

This shows that the system can adapt to varying environmental conditions and provide accurate irrigation control.

4.6 SOIL MOISTURE DRIVEN IRRIGATION PERFORMANCE

The irrigation system operates based on real-time soil moisture values, making it efficient and reliable. During testing, the soil moisture sensor accurately detected variations in water content and sent corresponding signals to the microcontroller. When the soil became dry, the system activated the water pump automatically, ensuring timely irrigation. As the soil moisture level increased, the system continuously monitored the changes and turned OFF the pump once the desired level was reached. This prevented over-irrigation and ensured optimal water usage. The switching process was smooth and did not show unnecessary fluctuations.

The system maintained consistent moisture levels, which is important for proper plant growth. By avoiding both excessive and insufficient watering, it supports better root development and improves crop health. The automation also reduces manual effort and ensures efficient resource utilization. Overall, the system demonstrated stable and accurate performance in controlling irrigation based on soil conditions, making it suitable for practical agricultural applications.

The system monitors environmental conditions such as temperature and humidity using the DHT11 sensor. These parameters are essential for understanding the surrounding climate and its effect on plant growth. The sensor provided continuous and stable readings, which were displayed on the LCD screen in real time. The observed values helped in analyzing how environmental conditions change during operation. Temperature variations can affect soil moisture through evaporation, while humidity influences the moisture content in the air. Monitoring these parameters provides additional insight into field conditions. Although the sensor has basic accuracy, it performed reliably for this application. The readings were consistent and sufficient for general monitoring purposes.

This feature enhances the overall functionality of the system by combining environmental data with soil moisture information. Thus, the system not only controls irrigation but also provides useful environmental data, making it more effective for smart agriculture applications.

4.7 RELAY-BASED PUMP CONTROL PERFORMANCE

The relay module is used to control the water pump based on signals from the microcontroller. It acts as a switching device that allows safe operation of the pump using low-power control signals. During testing, the relay responded quickly and accurately to changes in soil moisture levels.

When the moisture level dropped below the threshold, the relay activated the pump without delay. Similarly, when sufficient moisture was reached, the relay turned OFF the pump immediately. The switching operation was stable and consistent throughout the testing period.

The use of relay ensures proper isolation between the control circuit and the high-power pump, protecting the system from electrical damage. It also reduces manual effort by automating pump operation. The relay-based control mechanism proved to be efficient, reliable, and suitable for continuous use in irrigation systems.

4.8 REAL-TIME DATA VISUALIZATION ANALYSIS

The system provides real-time data visualization through the LCD display, allowing users to monitor system parameters easily. The display shows important values such as soil moisture, temperature, and system status. During operation, the data was updated continuously without delay. The information was clear and easy to read, making it simple for users to understand system behaviour. Users could quickly identify whether the soil was dry or wet and whether the pump was ON or OFF.

This real-time feedback improves usability and allows better monitoring of the system. It also helps in identifying any abnormal conditions quickly, enabling timely action. The display system enhances user interaction and makes the system more practical and user-friendly for agricultural use.

4.9 REMOTE DATA COMMUNICATION PERFORMANCE

The IoT functionality enables remote monitoring by transmitting sensor data through the Wi-Fi module. This allows users to access real-time information from any location with internet connectivity. During testing, the system successfully transmitted data with good accuracy. The received data matched the actual sensor readings, indicating reliable communication. The system provided consistent updates under stable network conditions.

Remote monitoring reduces the need for physical presence in the field and improves convenience for farmers. It allows quick access to system status and environmental conditions. Although performance depends on network availability, the system demonstrated effective data transmission. The integration of IoT adds significant value by transforming the system into a smart and connected agricultural solution.

4.10 LIMITATIONS OBSERVED DURING TESTING

During testing, some limitations were observed. Sensor readings may slightly vary depending on soil type and environmental conditions. This may affect accuracy in certain situations. The system also depends on proper power supply for continuous operation. Any interruption in power may affect performance. In addition, internet-based monitoring requires stable network connectivity. In areas with poor network coverage, remote monitoring may be affected.

CHAPTER 5

APPLICATIONS AND ADVANTAGES

5.1 REAL TIME APPLICATIONS

The Smart Agriculture Monitoring System has wide applications in various fields. It can be used in agricultural farms for automatic irrigation and monitoring of soil conditions.

It is also useful in greenhouses where environmental conditions need to be controlled carefully. The system can be applied in gardens, nurseries, and research fields for better plant growth.

In addition, the system can be used in smart cities for water management and environmental monitoring. Its flexibility makes it suitable for different types of applications.

5.2 ADVANTAGES

The system offers several advantages such as automation, efficiency, and ease of use. It reduces water wastage by providing irrigation based on actual soil conditions.

The system minimizes manual effort and saves time for farmers. It provides real-time data, which helps in better decision-making. The system is cost-effective and suitable for small and medium-scale farmers.

It also improves crop productivity by maintaining proper soil moisture and environmental conditions.

CONCLUSION

The Smart Agriculture Monitoring System using IoT has been successfully designed and implemented. The system is capable of monitoring important environmental parameters such as soil moisture, temperature, and humidity in real-time. Based on the soil moisture condition, the system automatically controls the irrigation process using a relay and water pump. This reduces manual effort and ensures efficient water usage.

The use of sensors and microcontroller makes the system reliable and accurate. The LCD display provides real-time data locally, and the Wi-Fi module enables remote monitoring. The system improves agricultural productivity by maintaining proper soil conditions and reducing water wastage. Overall, the project meets its objectives and provides a practical solution for smart farming.

The system can be further improved by adding advanced features and technologies. A mobile application can be developed to provide a better user interface for monitoring and control. Cloud storage can be integrated to store data and perform long-term analysis. Solar power can be used to make the system energy-efficient and suitable for remote areas.

Additional sensors such as soil pH sensor, water level sensor, and rainfall sensor can be included for more accurate monitoring. Advanced technologies like Artificial Intelligence and Machine Learning can be used for predictive analysis and smart decision-making. These enhancements can make the system more powerful and suitable for large-scale agricultural applications. The proposed system has wide applications in modern agriculture. It can be used in agricultural fields for automatic irrigation and monitoring. It is also suitable for greenhouses where controlled environmental conditions are required.

The project also highlights the importance of automation in reducing human dependency. Farmers no longer need to manually monitor soil conditions or operate irrigation systems continuously. This not only saves time and labor but also ensures that crops receive timely and appropriate care. The automation process helps maintain consistency in irrigation, which is essential for healthy crop growth.

Another important aspect of the system is its cost-effectiveness. The components used, such as Arduino, sensors, and Wi-Fi modules, are affordable and easily available. This makes the system suitable for small and medium-scale farmers who may not be able to invest in expensive technologies. The simplicity of the design also makes it easy to install and operate without requiring advanced technical knowledge.

The system also contributes to sustainable agriculture by promoting efficient use of resources. Water is supplied only when required, which helps in conserving water and preventing wastage. Proper irrigation also supports soil health and reduces the risk of overwatering or under-watering. This ensures long-term productivity and environmental balance.

Furthermore, the modular design of the system allows easy expansion and customization. Additional sensors and features can be integrated based on specific requirements.

The successful implementation of this project proves that IoT-based solutions can play a major role in improving agricultural practices. It provides a foundation for further research and development in the field of smart agriculture. With continuous improvements and adoption of advanced technologies, such systems can significantly enhance productivity and support the growing demand for food.

In conclusion, the Smart Agriculture Monitoring System using IoT is a reliable, efficient, and practical solution for modern farming. It not only meets the current requirements of agriculture but also provides opportunities for future enhancements. The system has the potential to contribute towards the development of intelligent and sustainable agricultural systems.

FUTURE SCOPE

The proposed Smart Agriculture Monitoring System can be further enhanced by incorporating advanced technologies and additional features. One of the major improvements can be the integration of Artificial Intelligence and Machine Learning algorithms. These technologies can analyze collected data and predict future irrigation requirements based on patterns and environmental conditions. Another important enhancement is the development of a mobile application or web-based interface. This will provide a user-friendly platform for farmers to monitor and control the system remotely. Notifications and alerts can be sent to users regarding soil conditions, pump status, and system performance.

The system can also be expanded by adding more sensors such as soil pH sensors, water level sensors, and rainfall sensors. These sensors will provide more detailed information about soil health and environmental conditions, enabling better decision-making. In future, the communication system can be improved by using technologies such as GSM, LoRa, or NB-IoT. These technologies provide better connectivity in rural and remote areas where Wi-Fi may not be available. This will make the system more reliable and widely usable.

The use of renewable energy sources such as solar power can make the system more energy-efficient and independent of external power supply. Solar-powered systems are especially useful in agricultural fields where electricity availability may be limited. Further improvements can include the integration of automatic fertilizer control systems. This will allow the system to supply nutrients to crops based on soil conditions, improving crop quality and yield.

In addition, the system can be scaled up for large agricultural fields by deploying multiple sensor units and connecting them through a centralized control system. Cloud storage and data analytics can be used to store and analyze large amounts of data for long-term planning and optimization. With these advancements, the system can evolve into a fully automated and intelligent farming solution. It has the potential to transform traditional agriculture into a modern, efficient, and technology-driven system that supports sustainable development and improved agricultural productivity.

Furthermore, the system can be enhanced by incorporating real-time decision support systems that assist farmers in taking immediate and accurate actions. By combining sensor data with intelligent algorithms, the system can provide recommendations such as optimal irrigation time, water quantity, and crop management practices. This reduces dependency on manual judgment and improves farming efficiency.

Another possible enhancement is the integration of advanced data visualization techniques. Instead of displaying raw data, the system can present information in the form of graphs, charts, and dashboards. This makes it easier for users to understand trends and patterns in environmental conditions. Visual representation of data helps in quick analysis and better decision-making.

The system can also be improved by introducing fault detection and self-diagnostic features. These features can identify issues such as sensor malfunction, communication failure, or power interruption. Once a fault is detected, the system can notify the user or take corrective action automatically. This improves system reliability and reduces maintenance efforts.

In addition, the use of edge computing can be considered for faster data processing. Instead of sending all data to the cloud, some processing can be done locally at the microcontroller level. This reduces latency and ensures quicker response to changing field conditions. It also minimizes dependency on continuous internet connectivity.

Security is another important aspect that can be enhanced in future developments. As the system involves data transmission over the internet, proper security measures such as data encryption and secure authentication can be implemented. This will protect the system from unauthorized access and ensure safe operation.

The system can further be integrated with automated irrigation scheduling techniques. Based on crop type, soil condition, and weather data, the system can generate optimized irrigation schedules. This improves water management and ensures that crops receive the right amount of water at the right time.

Another advancement can be the use of drone technology for monitoring agricultural fields. Drones can capture images and provide aerial views of crops, helping in identifying issues such as dry areas, pest attacks, or uneven irrigation. This information can be combined with sensor data for more accurate analysis.

The system can also support integration with smart farming equipment such as automated tractors and robotic systems. These technologies can work along with the monitoring system to perform tasks such as planting, fertilizing, and harvesting. This leads to a fully automated agricultural environment.

In future, the system can be customized for different types of crops and soil conditions. By adjusting parameters and thresholds, the system can be adapted to various agricultural requirements. This flexibility makes it suitable for diverse farming applications.

Overall, these enhancements will make the system more intelligent, efficient, and reliable. With continuous improvements, the Smart Agriculture Monitoring System can evolve into a comprehensive solution that supports modern agriculture and ensures sustainable development.

SOURCE CODE

```
#include <LiquidCrystal.h>

#define pumpPin1 6

#define pumpPin2 5

int temppin=A0;

const int soilmoisturePin=A1;

const int rs=13, en=12, d4=11, d5=10, d6=9, d7=8;

LiquidCrystallcd(rs, en, d4, d5, d6, d7);

void setup() {

  lcd.begin(16,2);

  pinMode(soilmoisturePin, INPUT);

  pinMode(pumpPin1, OUTPUT);

  pinMode(pumpPin2, OUTPUT);

  Serial.begin(9600);

}

void loop() {

  int potStatus= analogRead(soilmoisturePin);

  int val=analogRead(temppin);

  float mv=(val/1024.0)*5000;

  float t=mv/10;

  lcd.setCursor(0,0);

  lcd.print("Temp: ");

  lcd.print(t);
```

```

lcd.print(" C");

lcd.setCursor(0,1);

lcd.print("WATER LEVEL:");

lcd.print(potStatus);

Serial.print("Temp: ");

Serial.print(t);

Serial.println(" C");


if (potStatus<= 200) {

digitalWrite(pumpPin1, LOW);

digitalWrite(pumpPin2, HIGH);

Serial.print("Water level is LOW, Turn ON the Pump : ");

Serial.println(potStatus);

} else {

digitalWrite(pumpPin1, LOW);

digitalWrite(pumpPin2, LOW);

Serial.print("Water level is GOOD, Turn OFF the Pump : ");

Serial.println(potStatus);

}

delay(5000);

}

```

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